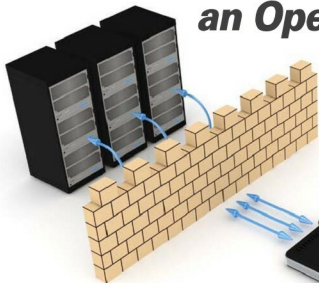


Get Your Hands on Vyatta— an Open Source Network Operating System



This article takes a look at Vyatta, a Debian-based Linux operating system that provides a software-based virtual firewall and VPN services for IPv4 and IPv6 networks.



Before going into the details of Vyatta, it is a good idea to revisit a few basics.

OSI layers: Although TCP/IP is the most widely used protocol suite today, as a rule of thumb, the explanation for various layers is based on the ‘Seven layers Open System Interconnect model’ – generally referred to as the OSI model. OSI layers were designed to segregate the various functions required to transfer data from one device to another over a network, thus enabling various equipment manufacturers to concentrate primarily on their area of expertise.

Imagine the transition from IPv4 to IPv6. Without the OSI, every equipment manufacturer or software developer would have to change their products based on the new addressing scheme. However, with OSI layers in place, change in the concerned equipment is required only in the network layer (Layer 3 of OSI)—the components operating at other layers do not require any change. For example, to use IPv6 instead of IPv4, no change is required in Physical Layer 1—copper, fibre cables; in Datalink Layer 2—Ethernet cards (MAC address); in Transport Layer 4—TCP or UDP; or even Application Layer 7—various protocols such as POP, SMTP, http, etc.

Router: If one wishes to travel from one place to another, there are various roads available for the journey.

But which path should one choose? Probably the one that is less congested and with a good road quality. In TCP/IP networks, the task of choosing the best path towards a destination is done by the router.

A router can communicate with two or more IP address ranges at the same time. Usually, a router IP address is the one that is set up as the default gateway in the computer system’s TCP/IP settings. If the destination IP address is not in the same subnet range as the source IP address, the packets are forwarded to the default gateway. The router (Default gateway) maintains a table of its interfaces and the corresponding IP addresses connected to them. Once a packet is received for a particular destination IP address, it is forwarded to the corresponding interface. (This explanation is for giving fair idea about router functionality especially to non technical readers / beginners.) Invariably, you will find a router connected where the Internet connection is shared between a number of computer systems and devices.